

Power Budget: Working Document

Digital Audio Player • Rev 6

Author: Alexander Loker. Design decisions, component selection, analysis and verification.

Drafted by: Claude (Anthropic). Structure and prose from the author's direction and source data.

Contents

1	Summary	1
2	Design Requirements	1
3	Method	1
4	Source Data	1
5	Conversion Efficiency	2
6	Operating Scenarios	2
6.1	S1: Standby	2
6.2	S2: Playback, display off (dominant)	3
6.3	S3: Browsing, display on	3
6.4	S4: Absolute peak (transient)	3
7	Battery Sizing	4
7.1	Selected Cell	4
7.2	Weighted Average and Derating	4
7.3	Result	4
8	Charging	5
9	Regulator and Thermal Sizing	5
10	Confidence Register	6
11	Validation Plan	6
12	Revision Log	6
13	Sources	7

1. Summary

This document converts per-block power figures into three numbers the design needs: an average draw for battery sizing, a peak draw for regulator and thermal sizing, and a standby draw for shelf life. These are different numbers and are kept separate.

Result	Value
Playback, display off (S2)	≈ 605 mW (164 mA at 3.7 V)
Weighted average (90% S2, 10% S3)	≈ 641 mW (173 mA)
Runtime on derated capacity	≈ 7.8 h against a 7 h target
Absolute peak (S4)	≈ 1.49 W (403 mA at 3.7 V)
Standby (S1)	≈ 11 mW (≈ 26 days to flatten the cell)

2. Design Requirements

Requirement	Value
Battery life	≥ 7 h continuous playback on derated capacity, 90% S2 / 10% S3 usage mix
Maximum audio output	0.6 V RMS (0.85 V peak) into the Simgot EW300 (28 Ω ±15%)
Cell	Jauch LP605060JU+PCM, 1S1P, 1850 mAh (6.8 Wh)
Charge current	500 mA (0.27C)
Constraints	Low noise · Small footprint

3. Method

Summing every block's active figure produces a number no operating state reaches. Three blocks dominate that error:

- The microSD card is active roughly 2% of the time. FLAC 16/44.1 stereo is about 90 kB/s; a Class 10 card sustains 10 MB/s. Its 330 mW active figure contributes about 7 mW averaged.
- The OPA1622's 185 mW is full scale (0.6 V RMS) into 23.8 Ω on both channels. At IEM listening levels the output stage draws little more than quiescent. Quiescent governs battery life; peak governs thermal and regulator sizing.
- The display is off during playback by design, putting its duty cycle near 5% in the dominant use case.

Conversion losses in each scenario use the loss ratio carried from the Rev 3 estimate (about 33% of load in S2, 28% in S3, 24% in S4). This is conservative and is replaced by per-branch figures as simulations complete.

4. Source Data

Block	Active	Passive	Rail	Basis
DAC (ES9038Q2M)	53 mW	5 mW	3.3 V	Datasheet p. 50 at DVCC = 3.3 V. 26 mW on +3V3_ANALOG (VCCA 2 mA, AVCC 6 mA), 26 mW on +3V3_DIG.
MCU (ESP32-S3)	165 mW	7 μW	3.3 V	Datasheet
microSD (32 GB Class 10)	330 mW	3 mW	3.3 V	Typical
Oscillator (one active)	7 / 18 mW	≈ 0	3.3 V	ECS 2.2 mA max / Abracon 5.5 mA max; disabled part in standby
Output amp (OPA1622, 2 ch)	185 mW max	5 μA/ch	±3.3 V	Full scale into 23.8 Ω, both channels, plus 34 mW quiescent (2.6 mA/ch)
I/V and summing (OPA1612, 6 ch)	143 mW	No shutdown	±3.3 V	3.6 mA/ch × 6 × 6.6 V; off only when the analog rails are off
Display + backlight (2.4 in)	300 mW	None	3.3 V + boost	Rough estimate
Charger (BQ24250) leakage	None	60 to 80 μW	BAT	Datasheet: 16 μA typ, 22 μA max, high-impedance mode

OPA1622 full-scale supply power (class AB, sine, per channel)

$$P_{\text{supply}} = \frac{2 V_S I_{\text{pk}}}{\pi} = \frac{2 \times 3.3 \text{ V} \times 35.7 \text{ mA}}{\pi} = 75 \text{ mW} \Rightarrow 2 \times 75 + 34 \approx 185 \text{ mW}$$

$$P_{\text{load}} = \frac{V_{\text{RMS}}^2}{R_L} = \frac{0.6^2}{23.8 \Omega} = 15 \text{ mW per channel}$$

Source: [OPA1622 Datasheet, SBOS727B \(Texas Instruments\)](#) (quiescent 2.6 mA per channel, shutdown 5 μA per channel).

Source: [ES9038Q2M Datasheet v1.4 \(ESS Technology\)](#), p. 50; [OPA1612 Product Page and Datasheet \(Texas Instruments\)](#); [ECS-2520MVLCDatasheet \(ECS Inc.\)](#); [ASTX-H11 Datasheet \(Abracon\)](#).

5. Conversion Efficiency

Branch	Topology	Efficiency	Note
Charger input to SYS	BQ24250, 3 MHz synchronous buck	90 to 93%	Datasheet curves at 5 V in, 500 mA
Digital 3.3 V	LTC3440 buck-boost, 600 kHz	≈ 90%	Switcher only
Analog +3.3 V	LTC3440 to 3.9 V, then LT3042	≈ 77%	≈ 90% × LDO ≈ 85%
Analog -3.3 V	LTC3440 to 3.9 V, LT3462A, then LT3093	≈ 52 to 55%	LT3462A simulated 67.7% at 98 mA, 71.8% at 27 mA. The LDO loss is deliberate: it buys the noise performance.
Backlight	Constant-current boost to the LED string	≈ 85%	Only when the display is on

6. Operating Scenarios

6.1 S1: Standby

Block	Draw	Basis
DAC	≤ 5 mW	Standby at DVCC = 3.3 V
MCU	< 0.1 mW	Deep sleep
microSD	0 mW	Power-gated by load switch
Analog branch (OPA1612, OPA1622, LDOs, inverter)	≈ 0 mW	LTC3440 3.9 V shut down; the inverter follows it
Oscillators	≈ 0 mW	OE low
Display	0 mW	Off
Charger leakage	≈ 0.08 mW	16 to 22 μA from BAT in high-impedance mode
Regulator quiescent	≈ 5 mW	Estimate; sum actual I_Q for the tree
Total S1	≈ 11 mW	Sets shelf life, not runtime

At 11 mW from a 6.8 Wh cell, standby alone flattens the battery in about 26 days before cell self-discharge. This depends on power-gating the SD card and shutting the analog branch down: the OPA1612 has no shutdown pin, so its 143 mW is only removed by turning its rails off. SYSOFF is not the standby mechanism; it disconnects the battery from the system entirely and is a shipping mode.

6.2 S2: Playback, display off (dominant)

Block	Duty	Effective draw	Basis
MCU	100%	165 mW	Decode and I2S service
DAC	100%	53 mW	Active
microSD	2% / 98%	≈ 10 mW	$330 \times 0.02 + 3 \times 0.98$
Oscillator	100%	≈ 18 mW	Abracon (48 kHz family), worst case
OPA1622 output (2 ch)	100%	≈ 50 mW	34 mW quiescent + ≈ 16 mW typical signal
I/V and summing (6 ch)	100%	143 mW	Quiescent
Display	5%	≈ 15 mW	Brief wakes
Load subtotal		≈ 454 mW	
Conversion losses		≈ 151 mW	33% loss ratio

Block	Duty	Effective draw	Basis
Total S2		≈ 605 mW	164 mA at 3.7 V

6.3 S3: Browsing, display on

Block	Effective draw	Basis
MCU	165 mW	Active
DAC + full analog chain	230 mW	DAC 53 + OPA1612 143 + OPA1622 quiescent 34; not muted during browsing
Oscillator	≈ 18 mW	Worst case
microSD	≈ 40 mW	Higher duty during directory traversal
Display + backlight	300 mW	Continuous
Conversion losses	≈ 208 mW	28% loss ratio
Total S3	≈ 961 mW	260 mA at 3.7 V

6.4 S4: Absolute peak (transient)

Block	Peak draw	Basis
MCU	165 mW	Active
DAC	53 mW	Active
microSD	330 mW	Read burst
Oscillator	18 mW	Worst case
OPA1622 output	185 mW	Full scale, 0.6 V RMS into 23.8 Ω, both channels
I/V and summing	≈ 155 mW	Quiescent plus signal current
Display + backlight	300 mW	Full brightness
Conversion losses	≈ 288 mW	24% loss ratio
Total S4	≈ 1.49 W	403 mA at 3.7 V

7. Battery Sizing

7.1 Selected Cell

Attribute	Value
Manufacturer / MPN	Jauch Quartz, LP605060JU+PCM+2 WIRES 70MM
DigiKey PN	1908-LP605060JU+PCM+2WIRES70MM-ND
Configuration	1S1P Li-Po, 3.7 V nominal, 3.0 to 4.2 V
Capacity	1850 mAh (6.8 Wh)
Envelope	63.0 × 51.0 × 6.6 mm (includes PCM and tab)
Termination	70 mm flying leads, 2 wires, no NTC
Lead time	28 weeks (stock available at selection)

7.2 Weighted Average and Derating

Weighted average (90% S2, 10% S3)

$$P_{\text{avg}} = 0.90 \times 605 \text{ mW} + 0.10 \times 961 \text{ mW} = 641 \text{ mW}, \quad I_{\text{avg}} = \frac{641 \text{ mW}}{3.7 \text{ V}} = 173 \text{ mA}$$

Factor	Multiplier	Rationale
Usable depth of discharge	0.90	Cutoff above the PCM threshold so the device shuts down gracefully. The PCM trips around 2.5 V, roughly 20% below where firmware should stop, so relying on it loses power mid-write.
End-of-life capacity fade	0.85	Li-ion retains ≈ 80 to 85% of nameplate at 300 to 500 cycles. Charging at 0.27C and the switching charger's low heat both favour this figure.
Low-temperature derating	0.95	Device carried outdoors
Combined	0.73	$1850 \text{ mAh} \times 0.73 = 1350 \text{ mAh}$ usable

7.3 Result

Condition	Current	Runtime
Weighted average, nameplate capacity	173 mA	10.7 h
Weighted average, derated capacity (working figure)	173 mA	7.8 h
With 20% design margin, derated capacity	208 mA	6.5 h
Continuous browsing (worst realistic case)	260 mA	5.2 h

The 7 h target is met at 7.8 h on derated capacity, which is the working figure. The 20% design-margin case (6.5 h) falls short and is tracked as a risk until the display is measured, since the 300 mW display estimate is the largest uncertainty. Reserves if needed: MCU clock reduction to 160 MHz, or lower analog rails (the OPA1612 minimum is $\pm 2.25 \text{ V}$).

8. Charging

Parameter	Value	Note
Charge current	500 mA (0.27C)	Bottom of the BQ24250's 500 mA to 2 A range; the datasheet's own design example uses this value.
Charge time	≈ 4.5 to 5 h	Including CV taper
Safety timer	9 h, set via I2C	The 6 h default leaves under an hour of margin; a required configuration step.
Charger dissipation	$\approx 160 \text{ mW}$	At $\approx 93\%$ efficiency
Junction rise	$\approx 5 \text{ }^{\circ}\text{C}$	θ_{JA} 32.9 $^{\circ}\text{C/W}$ (VQFN-24); regulation threshold 125 $^{\circ}\text{C}$
Linear equivalent (rejected)	$\approx 650 \text{ mW}$, $\approx 21 \text{ }^{\circ}\text{C}$	Conducted into a cell 2 mm away while it nears full charge; heat plus high state of charge is the worst case for calendar aging.

- Input current is not charge current. Power is conserved through the buck: 500 mA at 5 V in yields roughly 600 mA into a 3.7 V cell at $\approx 90\%$ efficiency.
- Concurrency is handled in hardware. PowerPath reduces charge current when system load exceeds the input limit, so the S4 peak does not collapse the rail or break termination.
- Input current limit is set by the EN1/EN2 GPIOs or over I2C. V_{IN_DPM} reduces input current to hold the supply up on a weak source or long cable.

Source: [bq2425x Product Page and Datasheet \(Texas Instruments\)](#).

9. Regulator and Thermal Sizing

These use S4 (peak), never the average.

Item	Result
Charger inductor	System load plus charge current plus half the ripple: $403 + 500 + 222 \text{ mA} \approx 1.13 \text{ A}$, against 2 A saturation
LT3042 (+3.3 V)	$\approx 35 \text{ mA}$ quiescent (OPA1612 21.6 + OPA1622 5.2 + DAC analog 8 mA) + 71.3 mA signal $\approx 106 \text{ mA}$ peak, against the 125 mA programmed limit (1.18 $\times$) and 200 mA rating
LT3093 (–3.3 V)	$\approx 27 \text{ mA}$ quiescent + 71.3 mA signal $\approx 98 \text{ mA}$ peak
LT3462A input	$\approx 147 \text{ mA}$ at 98 mA output (572 mW simulated at 3.9 V)
LTC3440 (SYS $\rightarrow$ 3.9 V)	$106 + 147 \approx 253 \text{ mA}$, about 1% over the 250 mA design current; switch limit 2.5 A min
LTC3440 (SYS $\rightarrow$ 3.3 V digital)	$\approx 260 \text{ mA}$ peak against the 300 mA design current
OPA1622 dissipation	Worst-case class AB dissipation occurs near $0.64 \times V_{\text{supply}}$ peak output (2.1 V peak at $\pm 3.3 \text{ V}$), above the 0.85 V design maximum, so the device sits on the favourable side of the curve.
SYS and BAT	Absolute maximum 5 V; everything downstream must respect this.

Load steps: SD read bursts and output-stage transients are step loads on their rails. The charger's loop changes from voltage mode in CC to current mode in CV, so SYS transient behaviour differs between charge phases.

10. Confidence Register

Item	Status	Action
Display + backlight (300 mW)	Rough estimate	Measure the selected 2.4 in panel at operating brightness. Largest single uncertainty.
OPA1612 quiescent (3.6 mA/ch)	Datasheet typical	Confirm the maximum at $\pm 3.3 \text{ V}$; six channels at stake.
OPA1622 quiescent and shutdown	Datasheet	2.6 mA/ch typ (3.3 mA max); 5 μA /ch shutdown.
ES9038Q2M (53 mW)	Datasheet	Confirm the power mode and oversampling setting in use.
Oscillator current	Datasheet maximum	Measure the running part.
microSD figures	Typical	Verify against an industrial card datasheet or measure with a sense resistor in the card's ground return.
SD duty cycle (2%)	Calculated	Validate by instrumenting buffer refill frequency.
Cell capacity (1850 mAh)	Datasheet	Check the discharge rate it is quoted at; the design draws $\approx 0.09\text{C}$.
Charger efficiency ($\approx 93\%$)	Datasheet curve	Verify at the actual operating point.
Conversion losses	Ratio method	LT3462A simulated; replace the loss ratios with per-branch figures.
Regulator quiescent (5 mW)	Estimate	Sum actual I_Q for every regulator in the tree.

11. Validation Plan

- Instrument battery current with a sense resistor in the cell return, logged over a full discharge. This supersedes every estimate here.
- Log firmware state alongside current so scenarios can be separated rather than inferred.

- Characterise the spare cell's discharge curve at operating current; the same data builds the voltage-based fuel-gauge lookup table.
- Measure a full charge cycle: current profile, time to termination, charger and cell temperature. Confirm the 9 h safety timer is never approached and thermal regulation never engages.
- Verify the TS network's actual trip temperatures; β error alone shifts the cool and warm crossings by about 3 °C.
- Measure each regulator branch on the assembled board through its test points.

12. Revision Log

Rev	Date	Change
3 to 5	Earlier	Maximum output 0.6 V RMS; OPA1622 peak 185 mW; LTC3440 at 3.9 V; negative branch changed to the LT3462A inverter; efficiency and confidence register updated.
6	2026-09-27	Reformatted. DAC 53 mW at DVCC 3.3 V; OPA1612 six channels 143 mW (arithmetic corrected); oscillator row added; OPA1622 quiescent and shutdown from the datasheet; standby corrected (analog branch shut down; OPA1612 has no shutdown); LT3462A simulated efficiency; runtime 7.8 h derated; LT3042 peak 106 mA; validation plan updated.

13. Sources

1. [ES9038Q2M Datasheet v1.4 \(ESS Technology\)](#)
2. [OPA1612 Product Page and Datasheet \(Texas Instruments\)](#)
3. [OPA1622 Datasheet, SBOS727B \(Texas Instruments\)](#)
4. [ECS-2520MVL Datasheet \(ECS Inc.\)](#)
5. [ASTX-H11 Datasheet \(Abracon\)](#)
6. [bq2425x Product Page and Datasheet \(Texas Instruments\)](#)
7. [LTC3440 Product Page and Datasheet \(Analog Devices\)](#)
8. [LT3042 Datasheet \(Analog Devices\)](#)
9. [LT3093 Datasheet \(Analog Devices\)](#)
10. [LT3462/LT3462A Datasheet \(Analog Devices\)](#)